

ANSWERS WITH EXPLANATION

- 1.
- Option (d) is correct.**

$$\begin{aligned}
 |2B[\text{adj}(3A)]| &= |2B| |\text{adj}(3A)| \\
 &= 2^3 |B| |3A|^{3-1} \\
 &= 8 |B| (3^3 |A|)^2 \\
 &= 8 |B| 3^6 |A|^2 \\
 &= 8 \cdot \frac{1}{729} \times 3^6 \left(\frac{1}{2\sqrt{2}} \right)^2 \\
 &= 8 \times \frac{1}{8} \\
 &= 1
 \end{aligned}$$

- 2.
- Option (b) is correct.**

$$\text{Given that } iz^3 + z^2 - z + i = 0$$

$$\text{Put } z = i$$

$$i^4 + i^2 - i + i = 0$$

$$\Rightarrow 1 - 1 - i + i = 0$$

Satisfy them

$$\begin{aligned}
 (|z| + 1)^2 &= (|i| + 1)^2 \\
 &= 2^2 = 4
 \end{aligned}$$

- 3.
- Option (d) is correct.**

Total four digit numbers start with 1.

$$= 3 \times 2 \times 1 = 6$$

$$\therefore \text{Repetition of digit} = \frac{6}{3} = 2$$

$$\text{Sum of digits in each place} = (0 + 4 + 5) \times 2 = 18$$

 $\therefore$ Sum of all four digit numbers start with 1

$$= 1 \times 6 \times 1000 + 18 \times 100 + 18 \times 10 + 18$$

$$= 7998$$

Total four digit numbers start with 4

$$= 3 \times 2 \times 1 = 6$$

$$\text{Repetition of digit} = \frac{6}{3} = 2$$

$$\text{Sum of digits in each place} = (0 + 1 + 5) \times 2 = 12$$

 $\therefore$ Sum of all four digit numbers start with 4

$$= 4 \times 6 \times 1000 + 12 \times 100 + 12 \times 10 + 12$$

$$= 25332$$

Total four digit numbers start with 5

$$= 3 \times 2 \times 1 = 6$$

$$\text{Repetition of digit} = \frac{6}{3} = 2$$

$$\begin{aligned} \text{Sum of digits in each place} &= (0 + 1 + 4) \times 2 \\ &= 10 \end{aligned}$$

Total four digit numbers start with 5

$$= 6 \times 5 \times 1000 + 10 \times 100 + 10 \times 10 + 10$$

$$= 31110$$

$$\text{Total sum} = 7998 + 25332 + 31110$$

$$= 64440$$

- 4.
- Option (a) is correct.**

Given that x, y and z are cube roots of unity

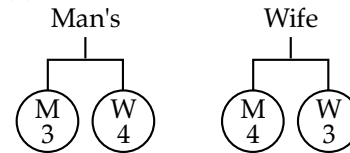
$$\therefore \text{Let } x = 1, y = \omega \text{ and } z = \omega^2$$

$$\text{Now, } xy + yz + zx = \omega + \omega^3 + \omega^2$$

$$= \omega + 1 + \omega^2 \quad (\because \omega^3 = 1)$$

$$= 0$$

- 5.
- Option (c) is correct.**



$$\text{Case-1 } 03 \quad 30 \rightarrow {}^3C_0 \times {}^4C_3 \times {}^4C_3 \times {}^3C_0 = 16$$

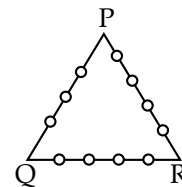
$$\text{Case-2 } 30 \quad 03 \rightarrow {}^3C_3 \times {}^4C_0 \times {}^4C_0 \times {}^3C_3 = 1$$

$$\text{Case-3 } 12 \quad 21 \rightarrow {}^3C_1 \times {}^4C_2 \times {}^4C_2 \times {}^3C_1 = 324$$

$$\text{Case-4 } 21 \quad 12 \rightarrow {}^3C_2 \times {}^4C_1 \times {}^4C_1 \times {}^3C_2 = 144$$

$$\text{Total number of ways} = 485$$

- 6.
- Option (a) is correct.**



Number of triangles

$$= {}^{12}C_3 - ({}^3C_3 + {}^4C_3 + {}^5C_3)$$

$$= 220 - (1 + 4 + 10)$$

$$= 205$$

- 7.
- Option (a) is correct.**

$$\log_b a = p$$

$$\Rightarrow a = b^p$$

$$\log_d c = 2p$$

$$\Rightarrow c = d^{2p}$$

$$\log_f e = 3p$$

$$\Rightarrow e = f^{3p}$$